

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Scenario Based Assessment
Weightage:	20% of CIA	Total Marks:	100
CO2 Statement:	<i>Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)</i>		
Student Name:	Y. Jithendra Reddy	Reg. No.:	192525303
Section / Batch:	2025	Date:	27-07-2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given scenario and identify the computational problem involved.
- Apply appropriate Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems wherever applicable.
- Clearly justify the chosen solution approach with proper reasoning and analytical interpretation.
- Demonstrate decision-making skills by comparing possible solutions and selecting the most suitable approach.
- Use diagrams, stepwise illustrations, or algorithmic representations wherever necessary.
- Maintain clarity, logical organization, and proper technical presentation throughout the answers.

Question Type	Focus Area	Questions
Scenario Based Assessment	Analyse real-world scenarios and apply Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems	Q1

SECTION A – SCENARIO BASED ASSESSMENT

Code of Conduct

I Y. Jithendra Reddy (192525303) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Y. Jithendra Reddy

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25%	Thoroughly understands the scenario with accurate interpretation of all requirements	Clearly understands the scenario with minor gaps	Basic understanding; some aspects of the scenario unclear	Poor understanding or misinterpretation of the scenario
Application of Concepts	30%	Applies appropriate concepts effectively to solve complex realworld problems	Applies relevant concepts correctly with minor errors	Applies basic concepts but with limited accuracy	Fails to apply appropriate concepts
Analytical & DecisionMaking Skills	20%	Demonstrates strong analysis and selects optimal solutions with justification	Good analysis with reasonable solutions	Limited analysis; solutions lack justification	Poor or no analysis; incorrect decisions
Solution Design & Approach	15%	Well-structured, innovative, and feasible solution approach	Logical and structured solution with minor gaps	Basic solution with limited structure or feasibility	Unclear or incorrect solution approach
Clarity, Justification & Presentation	10%	Clear, well-justified explanation with strong reasoning	Clear explanation with some justification	Limited clarity and weak justification	Poorly explained with no justification

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25%				

Application of Concepts	30%				
Analytical & Decision-Making Skills	20%				
Solution Design & Approach	15%				
Clarity, Justification & Presentation	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name: _____	Name: _____	Name: _____
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Q67. Navigation System – Closest Delivery Points Using Brute Force

Scenario

A navigation system must evaluate the distances between multiple delivery points in a city and determine the **closest pair of delivery locations** using the **Brute Force technique**.

The computational problem is a **Closest-Pair Problem**.

- Input: A set of delivery points with coordinates.
- Output: The pair of points having the **minimum distance**.
- Brute-force approach checks **every possible pair** and selects the smallest distance.

Example

Delivery Point Coordinates

P1 (1, 2)
P2 (4, 6)
P3 (5, 3)
P4 (8, 7)

The navigation system compares all pairs to find the nearest two delivery points.

a) Evaluation of All Pairs

For **n delivery points**, the algorithm compares each point with every other point exactly once.

Stepwise Evaluation

Step 1: Compare P1 with

- P2
- P3
- P4

Step 2: Compare P2 with

- P3
- P4

Step 3: Compare P3 with

- P4

Pair List

Pair Distance Formula

$$P1 - P2 \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$$

$$P1 - P3 \sqrt{((x_3 - x_1)^2 + (y_3 - y_1)^2)}$$

$$P1 - P4 \sqrt{((x_4 - x_1)^2 + (y_4 - y_1)^2)}$$

$$P2 - P3 \sqrt{((x_3 - x_2)^2 + (y_3 - y_2)^2)}$$

$$P2 - P4 \sqrt{((x_4 - x_2)^2 + (y_4 - y_2)^2)}$$

$$P3 - P4 \sqrt{((x_4 - x_3)^2 + (y_4 - y_3)^2)}$$

Brute Force Algorithm

min = ∞

for i = 1 to n-1

for j = i+1 to n

d = distance(P[i], P[j])

if d < min

min = d

closestPair = (P[i], P[j])

print closestPair

b) Time Complexity Derivation

For n points:

- First point $\rightarrow (n - 1)$ comparisons
- Second point $\rightarrow (n - 2)$
- Third point $\rightarrow (n - 3)$
- ...
- Last point $\rightarrow 0$

Total Comparisons

$$T(n) = (n - 1) + (n - 2) + \dots + 2 + 1$$

Using the sum of natural numbers:

$$T(n) = \frac{n(n - 1)}{2}$$

Complexity

$$T(n) = O(n^2)$$

Complexity Table

Case Time Complexity

Best Case $O(n^2)$

Average Case $O(n^2)$

Worst Case $O(n^2)$

Because every pair must be checked, the complexity remains **quadratic** in all cases.

c) Scalability Concerns

Comparison Growth

Number of Points Comparisons

10	45
100	4,950
1,000	499,500
10,000	49,995,000

Issues

1. Execution Time Increases

Large delivery networks require millions of distance calculations.

2. High CPU Usage

Each comparison involves:

- coordinate access,
- subtraction,
- squaring,
- addition,
- square-root computation.

3. Poor Real-Time Performance

For dynamic delivery systems, brute-force becomes too slow for:

- route optimization,
- live vehicle tracking,
- instant nearest-location suggestions.

4. Quadratic Growth

If the number of points doubles:

$$(2n)^2 = 4n^2$$

The work becomes **four times larger**, which limits scalability.

Better Alternative (Justification)

Method	Complexity
Brute Force	$O(n^2)$
Divide & Conquer	$O(n \log n)$
k-d Tree	$O(\log n)$ average query

Although brute force is **simple and accurate**, **Divide & Conquer** is more suitable for large-scale navigation systems.

Clarity, Justification & Presentation (10%)

Final Answer

a) All Pairs Evaluation

The navigation system evaluates every possible pair of delivery points using **two nested loops**, computes the Euclidean distance for each pair, and stores the pair with the **minimum distance**.

b) Time Complexity

Number of comparisons:

$$\frac{n(n-1)}{2}$$

Therefore,

$$O(n^2)$$

c) Scalability Concerns

- Comparisons grow **quadratically**.
- Execution time becomes very high for large numbers of delivery points.
- CPU and memory usage increase significantly.
- Hence, brute-force is **not scalable for large real-time navigation systems**.

Conclusion

The brute-force closest-pair approach is easy to implement and guarantees the correct result by examining **all possible pairs**. However, its **$O(n^2)$** time complexity makes it inefficient for large delivery networks. For real-world navigation systems handling thousands of delivery points, more efficient spatial searching techniques such as **Divide & Conquer or k-d Trees** should be preferred.

Weightage-Oriented Summary

Rubric Criterion	Coverage
Problem Understanding	✓ Closest-pair scenario identified
Application of Concepts	✓ Brute-force pair evaluation
Analytical Skills	✓ Complexity derived mathematically
Solution Design	✓ Structured algorithm provided
Clarity & Presentation	✓ Tables, formulas, and conclusion included